SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1025	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Problem Statement:	**Bus Ticket Reservation System**		
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Date:	10-10-2026	**Duration:**	60 minutes

Code of Conduct

I K Sampath Reddy (192424285) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *__K Sampath Reddy____*

Annexure A – Case Study Analysis

A Smart Bus Ticket Reservation System

Software Engineering Case Study: Problem Analysis, Solution Design and Methodology

No.	Section (as required in Annexure A)	Key content
1	Understand the Problem Statement	Problem, objectives, scope
2	Identify Key Stakeholders and Challenges	Roles, responsibilities, challenges
3	Analyze the Current Approach	Existing process, strengths, limitations, gaps
4	Propose a Suitable Solution	Modules, requirements, principles, diagrams
5	Justify the Chosen Methodology	Agile Scrum, comparison, sprint plan
6	Expected Outcomes and Limitations	Benefits, risks, limitations, future work

1. Understand the Problem Statement

1.1 The Problem in My Own Words

Millions of people travel by bus every day, but buying a ticket is still harder than it should be. Many passengers must visit a counter or call an agent, only to find that the seat they wanted is gone or that the information they were given was out of date. Operators suffer too: the same seat is sometimes sold twice, buses leave half-empty while other buses are full, and there is no clear picture of daily sales.

The problem statement asks for a **Bus Ticket Reservation System**: a software product that lets a passenger find a bus, see which seats are really free, pay, and receive a ticket in a few minutes, while giving the bus operator full control over routes, fares and reports. The key difficulty is not showing a list of buses. It is making sure that **one seat is never sold to two people**, even when many people are booking at the same moment.

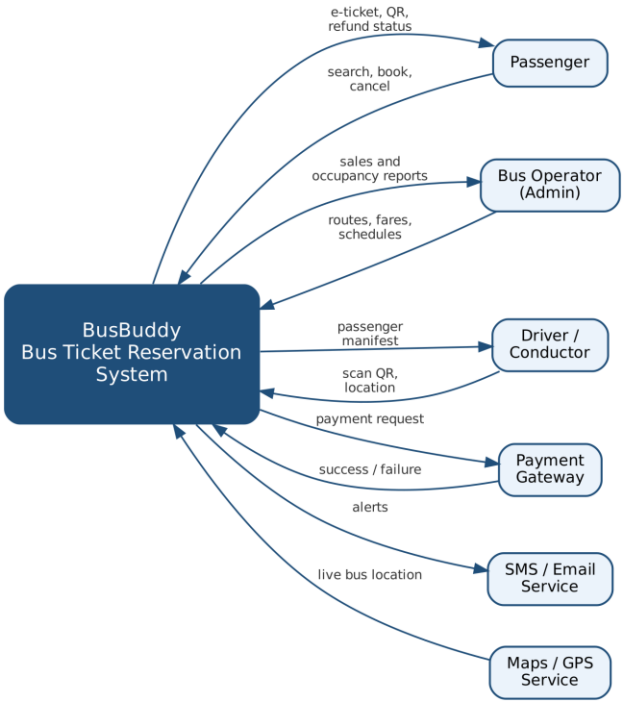


Figure 1: *Context diagram showing how BusBuddy interacts with people and external services*

1.2 Objectives of the Proposed System

#	Objective	What it means in practice
1	Easy search and booking	A passenger finds a bus and books a seat in under 3 minutes without visiting a counter.
2	Zero double booking	Every seat is locked for a short time while the passenger pays, so no two people can buy it together.
3	Secure online payment	Fares are paid through a trusted gateway; the system never stores card details.
4	Instant, paperless ticket	An e-ticket with a QR code is sent by SMS and email; the conductor simply scans it.
5	Fair cancellation and refund	Refunds follow clear time-based rules and are calculated automatically.
6	Better control for operators	Operators manage buses, routes, fares and see occupancy and revenue reports in real time.
7	Safe and transparent travel	Seat rules that protect women passengers, boarding-point choice and live bus tracking.

1.3 Scope of the Problem

The scope defines clearly what BusBuddy will and will not do, so that the project stays simple and finishable.

In Scope (what we build)	Out of Scope (not in this version)
• Passenger registration and login • Bus, route and schedule search • Live seat map with 8-minute seat lock • Online payment, e-ticket and QR code • Cancellation with automatic refund • Waitlist for fully booked buses • Admin panel for buses, routes, fares, reports • Conductor QR scanning and trip status • SMS / email notifications and live tracking	• Buying or maintaining the buses themselves • Driver payroll and vehicle maintenance • Hotel, cab or train booking • Fuel and GPS hardware manufacturing • International routes and multi-currency payments • Tax filing and government permit processing

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders, Roles and Responsibilities

A stakeholder is any person or organisation that uses the system, runs it, or is affected by it. Six main stakeholders were identified.

Stakeholder	Role and responsibilities	Major challenges faced today
Passenger	Searches buses, selects seats, pays, travels and cancels if plans change. Must give correct personal details.	Cannot see real seat availability; long queues; unclear refund rules; worried about safety and late buses.
Bus Operator / Admin	Owns the buses. Creates routes and schedules, sets fares and offers, reviews sales and manages cancellations.	Double booking; empty seats; paper records; no live sales data; difficult to change fares quickly.
Driver / Conductor	Checks tickets, boards passengers at the right stops and updates trip status (started, delayed, completed).	Manual ticket checking; fake or duplicate tickets; passenger lists are not up to date.
Payment Gateway Provider	Processes online payments and refunds and reports success or failure to the system.	Failed or duplicate payments; delayed refunds; strict security rules to follow.
Transport Authority	Sets rules for fares, safety and records; may audit trip and passenger data.	Needs accurate, tamper-free records; difficult to inspect paper registers.
Support and Development Team	Maintains the software, fixes defects, answers passenger complaints and releases new features.	Heavy complaint load; changing requirements; need for quick, safe updates.

2.2 Major Challenges and Issues

Looking across all stakeholders, the same issues appear again and again. They can be grouped into five themes:

- **Information gap:** passengers and operators do not see the same, up-to-date seat information.
- **Concurrency:** many users may try to book the last seat at the same time, leading to conflicts.
- **Trust and money:** payment failures, slow refunds and fake tickets reduce confidence.

3. Analyze the Current Approach

3.1 Existing System / Current Process

In many small and medium bus services, booking still works in a mostly manual way. A passenger goes to a counter or phones an agent. The clerk looks at a paper seating chart or a register, offers a seat, takes cash and writes a ticket by hand. Cancellation needs another visit to the counter and the refund is calculated by the clerk. Some operators use a basic website, but it often shows only a phone number or a static timetable and does not update seats in real time.

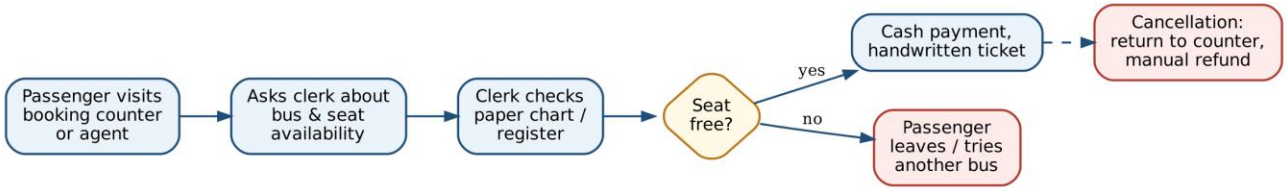


Figure 2: *Current manual booking and cancellation process*

3.2 Strengths and Limitations

Strengths of the current approach	Limitations of the current approach
Simple and familiar to staff and older passengers.	Passenger must travel to the counter; time-consuming.
Face-to-face help and personal trust.	Seat chart can be wrong or outdated; double booking is possible.
Cash payment needs no internet or bank account.	Handwritten tickets can be lost, copied or forged.
Very low technology cost to start.	Refunds are slow and calculated by hand, so disputes are common.
Works during power or network failures.	No sales reports; difficult to plan buses or change fares.
Staff can handle special requests directly.	Cannot scale to more routes, more buses or peak seasons.

3.3 Gap Analysis

The table below connects each gap found in the current process to its impact and to the way BusBuddy removes it.

Gap	Impact	How BusBuddy addresses it
No real-time seat view	Wrong information; lost sales	Live seat map updated for every booking and cancellation.
Double booking	Angry passengers; refunds	8-minute seat lock with database-level checks.
Manual, cash-only payment	Queues; risk of theft or mistakes	Secure online payment through a gateway.
Paper tickets	Forgery; lost tickets	E-ticket with unique QR code, validated on boarding.
Manual refunds	Delays and disputes	Automatic slab-based refund and status tracking.
No data or reports	Poor planning decisions	Admin dashboard with sales, occupancy and cancellation reports.
No safety or tracking features	Anxiety for passengers	Women-safe seat rule, boarding-point selection, live tracking and alerts.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Proposed Solution: BusBuddy

BusBuddy is a web and mobile-friendly bus ticket reservation platform. It follows a simple three-layer design: screens for users, independent business modules in the middle, and a database at the bottom. Beyond the basic booking features, it adds several ideas that make it different from a standard booking system:

Unique feature	Why it is useful
8-minute Smart Seat Lock	Selected seats are held for 8 minutes with a countdown. If payment is not completed, seats are released automatically, so no seat stays blocked and none is sold twice.
Women-Safe Seat Rule	Seats booked by a woman passenger are marked, and the system avoids placing an unrelated male passenger in the adjacent seat. This is configurable by the operator.
Waitlist Auto-Promotion	If a bus is full, passengers can join a waitlist. When a cancellation happens, the first waitlisted passenger is confirmed and notified automatically.
Boarding-Point Choice and Live Tracking	The passenger picks a boarding point and sees the bus on a map with an estimated arrival time, reducing waiting and confusion.
QR Boarding and Slab-Based Instant Refund	The conductor scans the QR code to verify the ticket. Refund is computed instantly from clear time slabs (see the refund slabs in Section 4.2).

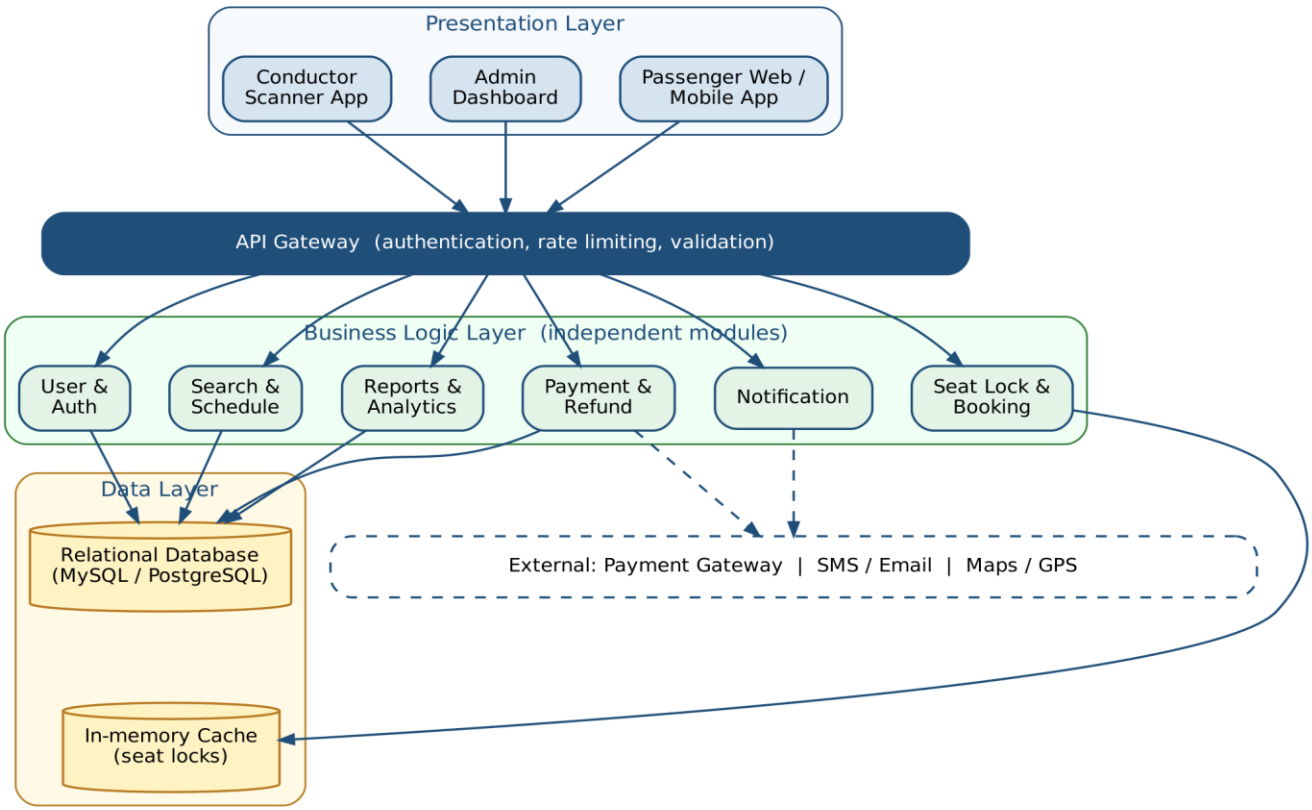


Figure 3: *Layered architecture of BusBuddy*

4.2 Major Functional Modules

The system is divided into six modules. Each module has one clear job and talks to the others only through defined interfaces.

Module	Main functions	Main users
1. User & Authentication	Registration, login, OTP verification, profile, role-based access	Passenger, Admin, Conductor
2. Search & Schedule	Search by source, destination and date; filters (price, time, bus type); route, bus and fare management	Passenger, Admin
3. Seat Lock & Booking	Live seat map, 8-minute lock, passenger details, booking confirmation, waitlist	Passenger
4. Payment & Refund	Gateway integration, payment status, cancellation rules, refund calculation	Passenger, Admin
5. Notification & Tracking	SMS and email alerts, e-ticket with QR, live location and delay alerts	Passenger, Conductor
6. Reports & Analytics	Daily sales, occupancy, cancellations, popular routes, export to PDF / Excel	Admin, Authority

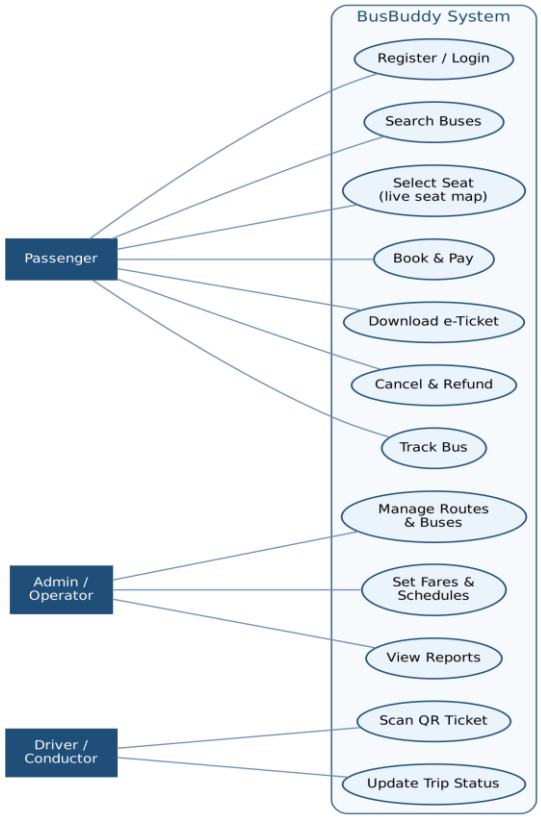


Figure 4: *Use case diagram of BusBuddy*

Reading the use case diagram

Each stick-box on the left is a user (actor). Each oval is something the system does for that user.

- **Passenger** has the most use cases, from searching to cancelling and tracking the bus.
- **Admin / Operator** configures the business: buses, routes, fares and reports.
- **Driver / Conductor** verifies tickets by scanning the QR code and updates the trip status.

Refund slabs (configurable)

More than 24 hours before departure: **90%** refund

12 to 24 hours: **75%** refund

4 to 12 hours: **50%** refund

Less than 4 hours: **no refund**

4.3 Functional and Non-Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement	Module
FR1	The system shall allow users to register, log in and reset passwords using OTP.	User & Auth
FR2	The system shall let passengers search buses by source, destination and date and filter the results.	Search
FR3	The system shall display a live seat map showing available, locked, booked and ladies seats.	Booking
FR4	The system shall lock selected seats for 8 minutes and release them automatically on timeout.	Booking
FR5	The system shall collect passenger details (name, age, gender) for every seat.	Booking
FR6	The system shall process online payments and confirm the booking only after payment success.	Payment
FR7	The system shall generate an e-ticket with a unique QR code and send it by SMS and email.	Notification
FR8	The system shall allow cancellation and calculate the refund using time-based slabs.	Payment
FR9	The system shall maintain a waitlist and promote the first passenger when a seat becomes free.	Booking
FR10	The admin shall be able to add, edit and remove buses, routes, schedules and fares.	Search / Admin
FR11	The conductor app shall scan and validate QR tickets and update trip status.	Notification
FR12	The system shall generate sales, occupancy and cancellation reports.	Reports

Non-functional requirements describe how well the system must work.

Category	Requirement	Measurable target
Performance	Search results and seat map load quickly, even during peak booking time.	Under 3 seconds for 95% of requests
Scalability	System handles growth in users, buses and routes without redesign.	5,000 concurrent users
Reliability	Bookings are never lost or duplicated; automatic recovery from failures.	99.5% uptime; zero double booking
Security	Passwords are hashed, traffic is encrypted and card data is never stored.	HTTPS / TLS; role-based access
Usability	Simple screens that work on mobile; booking in three clear steps.	New user books in under 3 minutes
Maintainability	Code is modular, documented and covered by automated tests.	At least 80% test coverage
Compatibility	Works on common browsers and Android / iOS phones.	Latest 2 versions of major browsers

4.4 Software Engineering Principles Incorporated

Principle	How it is built into BusBuddy	Example
Modularity	The system is split into six independent modules with clear interfaces; each can be built and tested alone.	Payment module can be changed without touching search.
Scalability	Stateless services, an in-memory cache for seat locks and database indexing allow more servers to be added.	Add servers during festival seasons.
Maintainability	Layered design, coding standards, code reviews, automated tests and clear documentation.	A new refund rule is a configuration change.
Reliability	Database transactions, unique seat constraint, payment retry and daily backups.	Two users cannot hold the same seat.
Usability	Clean mobile-first screens, colour-coded seat map, countdown timer, clear error messages.	Seat colours: green free, grey booked, pink ladies.
Security	Password hashing, OTP, encrypted traffic, input validation, role-based access and audit logs.	Only admins open the reports page.

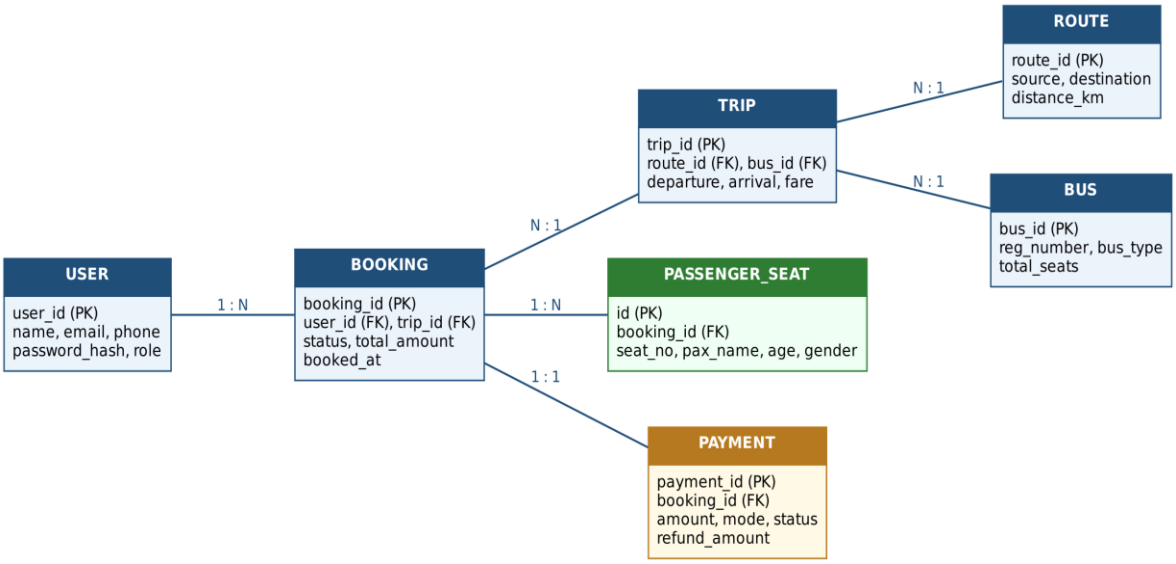


Figure 5: *Simplified entity-relationship diagram (database design)*

The database follows the rule that one seat on one trip can belong to only one active booking. This rule is enforced inside the database itself (a unique constraint on trip and seat number), which is the final safety net against double booking.

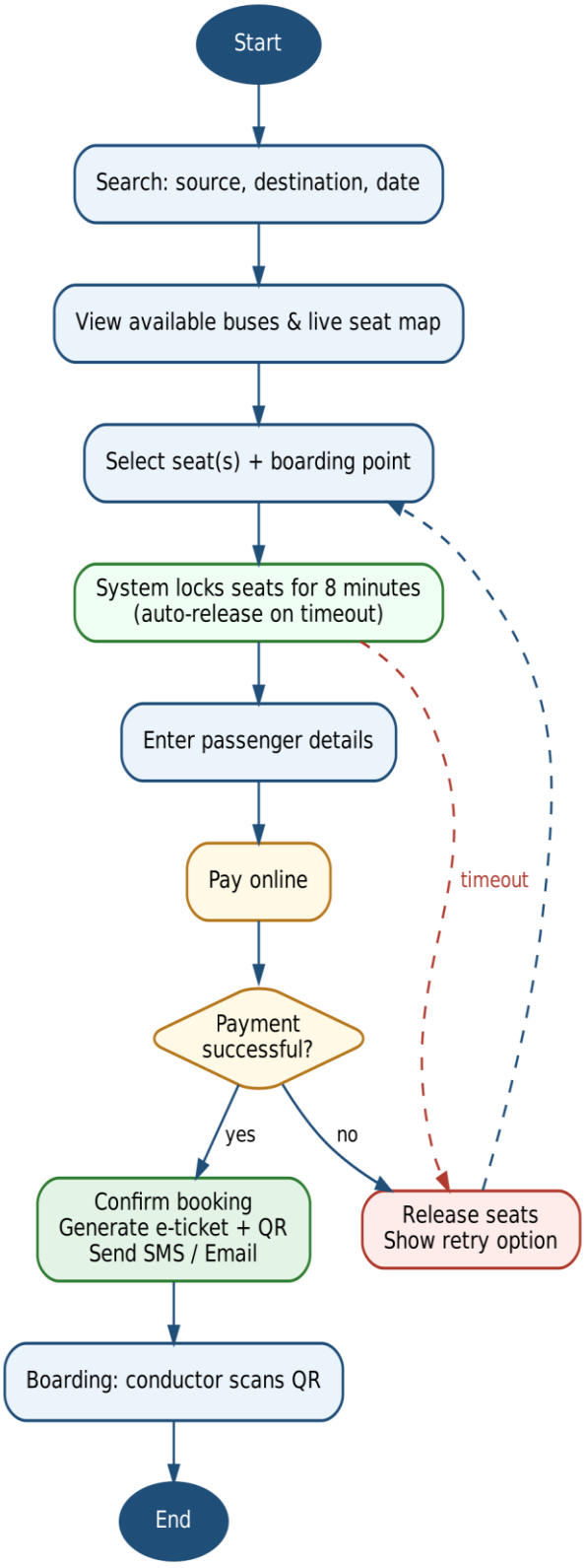


Figure 6: *Booking workflow with seat lock*

How a booking works, step by step

1. The passenger searches by source, destination and date.
2. The system shows buses and a live seat map.
3. The passenger selects seats and a boarding point.
4. The seats are **locked for 8 minutes**. A timer is shown.
5. Passenger details are entered and payment is made online.
6. If payment **succeeds**, the booking is confirmed, an e-ticket with QR is generated and an SMS / email is sent.
7. If payment **fails or the timer ends**, the seats are released and the passenger can retry.
8. At boarding, the conductor scans the QR code to verify the ticket.

> **Why this matters**
> The lock gives the passenger time to pay, and the timeout stops seats from being blocked forever. Together they remove double booking.

5. Justify the Chosen Methodology / Framework

5.1 Selected Model: Agile with the Scrum Framework

For BusBuddy, **Agile Scrum** is selected. Scrum builds the product in short cycles called sprints (two weeks each). At the end of every sprint a working, tested part of the system is delivered and shown to the stakeholders, who can then give feedback before the next sprint begins.

5.2 Why Scrum Is Suitable

Criterion	Waterfall	Spiral	Agile Scrum (selected)
Changing requirements	Difficult and costly	Handled, but heavy	Welcomed after every sprint
Early user feedback	Only at the end	After each loop	Every 2 weeks
Risk handling	Found late	Strong, but formal	Found early through frequent testing
Working software	At the end of the project	After several loops	After every sprint
Documentation / cost	Heavy documents	High analysis cost	Light and just enough
Fit for BusBuddy	Poor	Moderate	**Best fit**

The main reasons for choosing Scrum are:

- **Requirements will evolve.** Operators and passengers will ask for new offers, payment modes and route rules once they see the product.
- **Core features are risky and need early testing.** Seat locking and payment are the heart of the system; Scrum lets us build and test them in the first sprints.
- **Value is delivered early.** After Sprint 2 the operator can already search buses and see seat maps; after Sprint 3 real bookings can start.
- **Small, cross-functional team.** A team of four to six developers fits the Scrum model well.

Scrum role	Responsibility in the BusBuddy project
Product Owner	Represents the bus operator and passengers; maintains the product backlog and sets priorities.
Scrum Master	Guides the team, removes obstacles and ensures Scrum practices (daily stand-up, sprint review, retrospective) are followed.
Development Team	Designs, codes, tests and delivers the features of each sprint.

5.3 Sprint Plan (12-Week Roadmap)

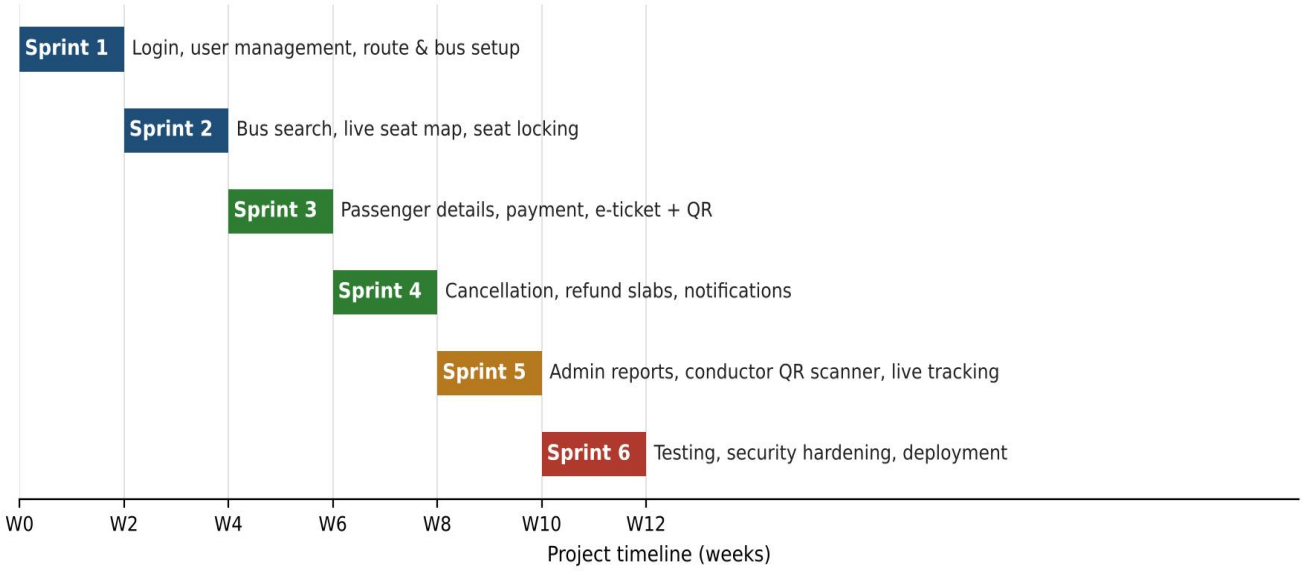


Figure 7: *Six two-week sprints with the main deliverable of each sprint*

5.4 Support for Successful Development and Maintenance

Scrum / Agile practice	How it helps the project
Product backlog and user stories	Keeps the work focused on real passenger and operator needs; features are ranked by value.
Daily stand-up (15 minutes)	Problems and blockers are found and solved the same day.
Sprint review and retrospective	Stakeholders give feedback; the team improves its process every two weeks.
Definition of Done and automated testing	A feature is complete only when coded, tested and reviewed, which keeps quality high.
Continuous integration and deployment (CI/CD)	Every change is built and tested automatically, enabling safe, frequent releases.
Maintenance as a backlog	Bugs, security patches and improvements are added to the backlog and planned like any other work.

6. Discuss Expected Outcomes and Limitations

6.1 Expected Benefits

Beneficiary	Expected benefit
Passengers	Book from anywhere in minutes; see real seats; get instant e-tickets and quick refunds; travel with more safety and confidence.
Bus operators	Higher seat occupancy; fewer errors and disputes; lower paperwork; real-time reports for better decisions.
Conductors / drivers	Fast ticket checking; accurate passenger list; no fake tickets.
Authorities	Digital, tamper-proof records that are easy to audit.

6.2 How the Solution Addresses the Identified Challenges

Challenge (Section 2)	Solution in BusBuddy
Information gap	Live seat map and real-time updates for everyone.
Concurrency and double booking	8-minute seat lock plus a database unique constraint.
Trust and money	Secure gateway, QR-verified tickets and automatic, rule-based refunds.
Safety and convenience	Women-safe seat rule, boarding-point choice, live tracking and delay alerts.
Operational load	Automation of booking, ticket checking and reporting reduces manual work.

6.3 Risks, Implementation Challenges and Limitations

Risk / challenge	Likelihood	Impact	Mitigation
Payment gateway downtime or failed payments	Medium	High	Retry option, backup gateway, automatic status check before releasing seats.
Peak load during festivals	Medium	High	Cache, load testing and the ability to add servers quickly.
Security attacks or data leaks	Medium	High	Encryption, input validation, regular security testing, minimum data storage.
Low adoption by older passengers and staff	High	Medium	Simple screens, staff training, keep counter booking linked to the same system.
Poor internet in rural areas	Medium	Medium	Lightweight pages, SMS fallback and offline QR check for conductors.
Changing requirements and scope creep	Medium	Medium	Prioritised backlog and sprint reviews.

Limitations of the proposed system: it depends on internet and a payment partner; live tracking needs GPS devices or a driver app on every bus; the women-safe seat rule uses the gender entered by the passenger and cannot verify it; and the first version covers only domestic routes in one currency.

6.4 Future Enhancements

- **AI-based demand prediction** to suggest dynamic fares and the best departure times.
- **Multi-language and voice booking** for wider accessibility.
- **Digital wallet, UPI auto-refund and loyalty points** for repeat travellers.
- **Integration with metro, rail and cab services** for end-to-end journey planning.
- **Emergency (SOS) button and trip sharing** with family during travel.
- **Mobile app with offline tickets** for areas with weak connectivity.

Conclusion

BusBuddy replaces a slow, error-prone and manual booking process with a simple, secure and scalable digital system. Built through Agile Scrum in short, testable sprints, it delivers value early, adapts to feedback and remains easy to maintain, while its unique seat lock, women-safe seating and instant refund features make travel easier and safer for everyone involved.